Final project proposal

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```
library(haven)
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.4.3
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
gss <- read_dta("gss7224_r3a.dta") %>%
  rename_with(tolower)

table(gss$year)
```

```
1972 1973 1974 1975 1976 1977 1978 1980 1982 1983 1984 1985 1986 1987 1988 1989
1613 1504 1484 1490 1499 1530 1532 1468 1860 1599 1473 1534 1470 1819 1481 1537
1990 1991 1993 1994 1996 1998 2000 2002 2004 2006 2008 2010 2012 2014 2016 2018
1372 1517 1606 2992 2904 2832 2817 2765 2812 4510 2023 2044 1974 2538 2867 2348
2021 2022 2024
4032 3544 3309
```

```r
gss_sub <- gss |> filter(year %in% c(2010, 2012, 2014, 2016, 2018, 2021, 2022, 2024))

table(gss_sub$year, gss_sub$happy, useNA = "ifany")
```

```
      1    2    3 <NA>
2010  538 1184  317    5
2012  593 1094  277   10
2014  786 1403  341    8
2016  806 1601  452    8
2018  701 1307  336    4
2021  783 2308  923   18
2022  779 1942  799   24
2024  684 1892  705   28
```

```r
table(gss_sub$year, gss_sub$partyid, useNA = "ifany")
```

```
      0   1   2   3   4   5   6   7 <NA>
2010 348 348 265 360 197 277 184  49   16
2012 356 343 235 373 157 250 192  54   14
2014 419 406 337 502 249 292 245  62   26
2016 463 496 405 473 292 364 270  72   32
2018 379 352 307 414 259 272 255  77   33
2021 822 541 471 817 327 384 524 114   32
2022 595 451 400 835 330 361 431 106   35
2024 532 372 340 829 274 378 460  83   41
```

```r
gss_sub |>
  summarise(across(c(age, sex, race, educ, marital, realinc, region),
                   ~ mean(is.na(.x))))
```

```
# A tibble: 1 x 7
     age     sex    race    educ marital realinc region
   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>  <dbl>
1 0.0298 0.00578 0.00759 0.00569 0.00207   0.110      0
```

```r
gss_chk <- gss_sub |>
  filter(partyid %in% 0:6, happy %in% 1:3) |>
  mutate(
    party3 = case_when(partyid <= 2 ~ "Dem",
                       partyid == 3 ~ "Ind",
                       TRUE ~ "Rep"),
    inc_miss = is.na(realinc)
  )

#
gss_chk |> group_by(year) |> summarise(miss = mean(inc_miss))
```

```
# A tibble: 8 x 2
  year        miss
  <dbl+lbl>  <dbl>
1 2010      0.110
2 2012      0.107
3 2014      0.0835
4 2016      0.0924
5 2018      0.0779
6 2021      0.123
7 2022      0.119
8 2024      0.111
```

```r
#   ×
gss_chk |> group_by(year, party3) |> summarise(miss = mean(inc_miss), .groups = "drop")
```

```
# A tibble: 24 x 3
   year      party3   miss
   <dbl+lbl> <chr>   <dbl>
 1 2010      Dem    0.0918
 2 2010      Ind    0.169
 3 2010      Rep    0.105
 4 2012      Dem    0.0880
 5 2012      Ind    0.141
 6 2012      Rep    0.116
 7 2014      Dem    0.075
 8 2014      Ind    0.124
 9 2014      Rep    0.0701
10 2016      Dem    0.0829
# i 14 more rows
```

```
#       very happy
gss_chk |> group_by(inc_miss) |> summarise(very_happy = mean(happy == 1))
```

```
# A tibble: 2 x 2
  inc_miss very_happy
  <lgl>         <dbl>
1 FALSE         0.251
2 TRUE          0.253
```

```
gss_sub |>
  group_by(year) |>
  summarise(
    n            = n(),
    wtssps_ok    = mean(!is.na(wtssps)),
    wtssnrps_ok  = mean(!is.na(wtssnrps)),
    wtssall_ok   = mean(!is.na(wtssall)),
    wtssps_mean  = mean(wtssps, na.rm = TRUE),
    wtssnrps_mean = mean(wtssnrps, na.rm = TRUE)
  )
```

```
# A tibble: 8 x 7
  year          n wtssps_ok wtssnrps_ok wtssall_ok wtssps_mean wtssnrps_mean
  <dbl+lbl> <int>     <dbl>       <dbl>      <dbl>       <dbl>          <dbl>
1 2010       2044         1           1          1       1.000          1.000
2 2012       1974         1           1          1       1.00           1.00
3 2014       2538         1           1          1       1.00           1.00
4 2016       2867         1           1          1       1.000          1.000
5 2018       2348         1           1          1       1.00           1.00
6 2021       4032         1           1          0       1.000          1.00
7 2022       3544         1           1          0       1.00           1.000
8 2024       3309         1           1          0       1              1.00
```

```
gss_sub |>
  summarise(
    cor_w   = cor(wtssps, wtssnrps, use = "complete.obs"),
    n_diff  = sum(abs(wtssps - wtssnrps) > 1e-6, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 2
  cor_w n_diff
```

```
  <dbl>  <int>
1 0.974  18624
```

```
gss_sub |>
  group_by(year) |>
  summarise(
    cor_w  = cor(wtssps, wtssnrps, use = "complete.obs"),
    pct_diff = mean(abs(wtssps - wtssnrps) > 1e-6, na.rm = TRUE)
  )
```

```
# A tibble: 8 x 3
  year      cor_w pct_diff
  <dbl+lbl> <dbl>    <dbl>
1 2010      0.973        1
2 2012      0.983        1
3 2014      0.976        1
4 2016      0.971        1
5 2018      0.965        1
6 2021      1.000        0
7 2022      0.974        1
8 2024      0.962        1
```

```
gss_sub |>
  group_by(year) |>
  summarise(vpsu_ok = mean(!is.na(vpsu)), vstrat_ok = mean(!is.na(vstrat)))
```

```
# A tibble: 8 x 3
  year      vpsu_ok vstrat_ok
  <dbl+lbl>   <dbl>     <dbl>
1 2010            1         1
2 2012            1         1
3 2014            1         1
4 2016            1         1
5 2018            1         1
6 2021            1         1
7 2022            1         1
8 2024            1         1
```

```
gss_sub |>
  group_by(year) |>
  summarise(n_strata = n_distinct(vstrat),
```

```
            n_psu    = n_distinct(vpsu),
            n_cells  = n_distinct(paste(vstrat, vpsu)))
```

```
# A tibble: 8 x 4
  year      n_strata n_psu n_cells
  <dbl+lbl>    <int> <int>   <int>
1 2010           134     2     268
2 2012            66     2     132
3 2014            65     2     130
4 2016            65     2     130
5 2018            78     2     156
6 2021             9  3563    3563
7 2022            66     2     132
8 2024            67     2     134
```